Assignment #3. Responsive Web Design (Media Queries + Bootstrap Grid)

Objective:
By the end of this assignment, students will be able to:

- make a layout responsive by hand with mobile-first media queries;
- build a responsive layout with the Bootstrap grid system: containers, rows, columns, breakpoints and ordering;
- use Bootstrap components and utility classes, and integrate Bootstrap with their own CSS;
- compare two solutions of the same layout problem and explain which one fits better and why.

In this assignment you build **one responsive page** on any topic you like, with **at least 4 sections** (for example: hero, about, services, gallery, contact).

You make it responsive in two ways: one section by hand with your own media queries (Task 1), and the rest of the page with Bootstrap (Task 2). Then you compare the two ways (Task 3).

Rules for Submission:

1. This is an individual assignment. If the works of different students are the same, the grade is **0**.
2. Submit before the deadline. **Late works are not accepted!** To get a grade you must **submit before the deadline AND defend your work at the practice lesson**. Submission without a defense will result in a grade of 0.
3. You must submit: your report in **Assignment3_YourName.docx** (or .pdf) format. If you submit in other formats, your grade will be **0**. All source code must be uploaded to the GitHub repository.
4. If you cannot explain your own code, the whole assignment is graded 0.
5. IN THE REPORT MUST BE:
   - Your name and group
   - Every task with screenshots and a short explanation in your own words
   - Screenshots of the whole page at different screen sizes (375px, 768px and 1280px)
   - A short conclusion about what you learned
   - The **GitHub repository URL** with the source code. Submissions without the link will receive a grade of 0

Setup:

1. Connect Bootstrap 5.3 from the CDN (copy the links from https://getbootstrap.com/docs/5.3/getting-started/introduction/):
   — the Bootstrap CSS **<link>** in the **<head>**;
   — the Bootstrap JS bundle **<script>** before **</body>** (needed for the navbar menu button, accordion, carousel, modal).
2. Connect your own **style.css after** the Bootstrap CSS link, so your styles can override Bootstrap.
3. Make sure the **<head>** has the viewport meta tag (see the Week 1 slides). Without it, media queries and the Bootstrap grid do not work on phones.

Task 1. Media queries by hand (one section)

Choose one section with repeated items (services, features, team, prices) and make it responsive with your own CSS only.

- **Use:** your own CSS (Grid or Flexbox) and **@media (min-width: ...)**. No Bootstrap classes, fixed widths or absolute positioning in this section.
- **Should work:**
  (1) Mobile-first: base styles are for a phone, media queries add changes for wider screens.
  (2) At least 2 breakpoints: 1 item per row on a phone, 2 on a tablet, 3 or more on a laptop.
  (3) At least 1 breakpoint where something else changes too: heading size, image position, spacing or alignment.

Task 2. Bootstrap grid and components (the rest of the page)

Build all other sections of the page with Bootstrap.

- **Use: container**, **row**, **col-*** with breakpoints, **order-***, **d-***, utility classes (**m-***, **p-***, **d-flex**, **text-***), a navbar and at least 1 more component (cards, accordion, carousel, modal or tabs). No floats, fixed widths or absolute positioning for the layout.
- **Should work:**
  (1) At least 1 section with cards: 1 per row on a phone, 2 on a tablet, 3 or more on a laptop.
  (2) At least 1 section with 2 or more columns (for example, text and image) that stack on a phone, with the more important one first.
  (3) At least 1 element that is hidden on a phone or shown only on a phone.
  (4) The navbar collapses into a menu button on small screens and links to the sections.

Task 3. Comparison of the two ways

Compare how you made the page responsive in Task 1 (media queries by hand) and in Task 2 (Bootstrap grid).

- **In the report:** a table with at least 3 rows (for example amount of code, control over design, speed of work, ease of changes), and 2–3 sentences on when you would choose each way.

Assessment criteria

Criterion	What is checked	Points
Task 1. Media queries	- the section is mobile-first and works as described at 375px, 768px and 1280px - the breakpoints change the layout, not only the sizes - desktop-first styles with max-width or Bootstrap classes in this section give half of the points at most	**10**
Task 2. Bootstrap grid and components	- the grid, the navbar and the component work as described at all 3 widths - no horizontal scroll at any width	**10**
Task 3. Comparison	- the table compares Task 1 and Task 2 with examples from this page, not general phrases - the choice of each way is explained	**5**
Report	- contains everything listed in the rules, in the order of the tasks - screenshots can actually be read, explanations are in the student's own words	**5**

Criterion	What is checked	Points
Defense	- explains the work the student has done and demonstrates the pages working live without errors (10) - answers the instructor's questions about their own code and the topic (30) - makes a change requested by the instructor without assistance or hints (30)	**70**
Total		**100**

Resources:

1. Abitova G.A. Web technologies Front-End Development. Part 2 (2022).
2. https://www.w3schools.com/css/css3_mediaqueries.asp
3. https://getbootstrap.com/docs/5.3/getting-started/introduction/
4. https://getbootstrap.com/docs/5.3/layout/grid/
5. https://www.w3schools.com/bootstrap5/
6. https://learn.astanait.edu.kz

***Questions will be based on the provided resources**

Good luck!